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20 Years of the Double Pukelsheim System in Switzerland: What Is the Mechanical Effect of the Transition from the Hagenbach-Bischoff Method on the Seat Distribution of Smaller Parties?

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1 Abstract

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2 List of Abbreviations

AG	Canton of Aargau
BGE	Decisions of the Swiss Federal Supreme Court (official collection)
CVP	Christian Democratic People's Party
DP	Double Pukelsheim system
FSC	Swiss Federal Supreme Court
GLP	Green Liberal Party
GR	Canton of Graubünden
HB	Hagenbach-Bischoff method
NW	Canton of Nidwalden
pp	percentage points
PR	proportional representation
SH	Canton of Schaffhausen
SVP	Swiss People's Party
SZ	Canton of Schwyz
UDC	Democratic Union of the Centre (French-language abbreviation for SVP)
UR	Canton of Uri
VS	Canton of Valais
ZG	Canton of Zug
ZH	Canton of Zurich

3 Introduction

On the 18 December 2002 the Swiss Federal Supreme Court (FSC) came to a conclusion, which disrupted the Swiss Electoral System. In BGE 129 I 185 the FSC criticized that in the elections for the Council of the City of Zurich, the thresholds for parties to win a seat varied heavily from district to district. Further did some districts have thresholds for parties to win a seat of up to 34%. Consequently, votes did not have equal effect, which undermines the electoral equality included in Art. 8 (Equality before the law) and Art. 34 (political rights) of the Federal Constitution. Based on this criticism the court decided that the City of Zurich had to implement corrective measures, but could choose their form (BGE 129 I 185 E. 7.3, E. 7.6.2–7.6.3 and E. 8.3.2; Schweizerische Eidgenossenschaft, 1999, Arts. 8 and 34). The Government of the Canton of Zurich decided to keep the constituency structure (Regierungsrat des Kantons Zürich, 2014, pp. 3–4). Instead the canton of Zurich commissioned the german mathematician Friedrich Pukelsheim to develop a comprehensive and at the same time minimally invasive solution for the problem. Eventually Pukelsheim came up with a method called "biproportional divisor method with standard rounding" which is today often simply called "Double Pukelsheimer" or "Biproportional apportionment". (Lüscher, 2024, pp. 9-10)

The distinguishing feature of the Pukelsheimer Proporz, is that it through the so called upper-apportionment artificially creates a single electoral district for the initial allocation of the parliamentary seats to the parties or list groups. Through this mechanism, the difference between seat and voteshares gets minimized and with that also the effective threshold for parties to enter the parliament gets lowered. (Lüscher, 2023, p. 575)

Through its upper-apportionment mechanism and the lowered threshold, the reform is especially relevant for the representation of smaller parties, since votes that would previously have been ineffective in small constituencies can now contribute to a seat gain at the cantonal level. Sandro Lüscher illustrates this mechanical effect with the case of Schwyz in 2016: under the old Hagenbach-Bischoff system, the Green Liberal Party would not have won a single seat, whereas under the new method they obtained three seats (Lüscher, 2025, p. 1-3).

Since Zurich first used the system in 2007, eight further cantons have used it: Schaffhausen (2008), Aargau (2009), Nidwalden (2014), Zug (2014), Schwyz (2016) , Valais (2017) , Uri (2020), Graubünden (2022). In 2025 also the voters from Basel Landschaft decided to implement the Biproportional apportionment from 2027 on (Lüscher , 2024, p.17 ; Kanton Basel-Landschaft, 2025a, p.7 ; Kanton Basel-Landschaft, 2025b) While nine of these cantons already implemented the algorithm, more cantons are currently discussing the implementation too. In Thurgau, the initiative "Mehr Transparenz bei Wahlen im Thurgau" is currently pending at the Grand Council of the Canton and a special parliamentary committee is currently examining the iniatitive, also in the canton of Lucerne, the iniatitive "Faire Wahlen ohne Listenflut" was submitted and the voters will be able to decide on the effectuation (Grosser Rat des Kantons Thurgau, n.d.; Regierungsrat des Kantons Luzern, 2026, pp. 19–20). In the canton of St.Gallen in 2026 an initiative to adapt the algorithm was launched (Initiativkomitee «Damit jede Stimme zählt», n.d.).

The amount of cantons who have already implemented the system or are debating about doing so, suggets that the decision by the FSC had a substantial impact on how Swiss citizens elect their regional parliaments. It seems that the double-Pukelsheimer through its wide implemen-

tation across cantons is a success story. Since the first election under double Pukelsheim in Zurich in 2007, the system has been used in 32 cantonal parliamentary elections.[1]

Taken together, this number of elections and almost 20 years of experience provide a reason to evaluate the impact of the reform, especially on smaller parties. How much has the reform changed smaller parties' seat shares and how large is the actual difference in seat allocation between the Hagenbach-Bischoff and Double Pukelsheim system? Are there other major differences, for example in how the reform performs in smaller versus larger parliaments, and how do smaller parties' seat shares and gains evolve across successive elections?

Existing research primarily examines Double Pukelsheim in legal, normative, or system-level terms. A few studies also conduct counterfactual comparisons, but these cover either individual elections or only one election per canton (Pukelsheim & Schuhmacher, 2011, pp. 1590–1592; Leuzinger, 2018, p. 42; Lüscher, 2024, p. 13). Although their results indicate that smaller parties can benefit from Double Pukelsheim, the magnitude and consistency of this mechanical allocation difference across successive elections remain less systematically examined.

This thesis addresses this gap by applying a consistent Hagenbach–Bischoff counterfactual to 30 elections in eight transition cantons. Two Graubünden elections are presented as contextual comparisons but excluded from the main transition estimate. Holding the observed post-reform votes and list structure constant, the analysis measures the difference in the share of seats won by smaller electoral lists and examines variation across institutional settings and successive elections.

4 Theory and Literature Review

4.1 Institutional and Theoretical Background

This chapter provides the institutional and theoretical background for analyzing the transition from Hagenbach-Bischoff to Double Pukelsheim and reviews the relevant literature. It first outlines the historical development of electoral systems in Switzerland and then explains the functioning and limitations of the Hagenbach-Bischoff and Double Pukelsheim methods. It subsequently examines the adoption and canton-specific implementation of Double Pukelsheim. Finally, the literature review situates this thesis within previous research before the expected effects of the transition are derived.

4.1.1 Historical Development

Until the late nineteenth century swiss voters elected the regional parliaments and great councils only under majority voting. In a majority voting system, the candidates or parties who receive the relative or absolute majority receive the seats, therefore this systematically favours the dominant parties. The disadvantages for smaller parties let some of the political

[1]The 32 elections comprise Zurich (2007, 2011, 2015, 2019, and 2023; Amt für Statistik und Daten Kanton Zürich, 2026); Schaffhausen (2008, 2012, 2016, 2020, and 2024; Kanton Schaffhausen, n.d.; Staatskanzlei Kanton Schaffhausen, 2020, 2024); Aargau (2009, 2012, 2016, 2020, and 2024; Department für Finanzen und Ressourcen des Kantons Aargau, 2009; Regierungsrat des Kantons Aargau, 2013; Statistisches Amt Kanton Aargau, n.d.); Nidwalden (2014, 2018, 2022, and 2026; Staatskanzlei Nidwalden, 2014, 2018, 2022, 2026); Zug (2014, 2018, and 2022; Kanton Zug, 2022a, 2022b, 2022c); Schwyz (2016, 2020, and 2024; Urteil des Bundesgerichts 1C_407/2011, 1C_445/2011 and 1C_447/2011 vom 19. März 2012); Valais (2017, 2021, and 2025; Kanton Wallis, 2016, 2020, 2024; Staat Wallis, 2017); Uri (2020 and 2024; Standeskanzlei Uri, 2024a, 2024b); and Graubünden (2022 and 2026; Kanton Graubünden, 2022, 2026).

minorities demand, to change to a proportional electoral system. In 1891 the canton of Ticino, was the first one to use a PR-System to vote for their Grand Council, till the end of World War I , 7 1/2 more cantons had followed. When the socialist and conservative minorities on their third attempt in 1918 achieved, that the National Council will be elected through a proportional system, it boosted the general adoption of it across Switzerland. Most of the cantons Grand Councils aswell as the National Council were then elected with the Hagenbach-Bischoff proportional system. In the beginning of the 21th century, the cantons of Graubünden and Appenzell Innerrhoden were the only ones to still vote their parliaments with the majority system. (Gilg, 2014, "Mehrheits- und Verhältniswahl" section ; Bundeskanzlei, 2013, p. 2) Nowadays the Kanton Appenzell Innerrhoden is the only remaining canton still voting the Grand Council by majority voting (Bundeskanzlei, 2025, p. 18). It can therefore be said, that proportional systems are the dominating electoral systems for cantonal parliaments in Switzerland.

In the last 20 years the prior dominance of the Hagenbach-Bischoff system has been demolished by the wide implementation of the DP. As of 2025, eleven cantons distribute the parliamentary seats with Hagenbach-Bischoff and ten cantons implemented the DP, two cantons namely Glarus and Basel-Landschaft use the Sainte-Laguë method and two namely Ticino and Vaud use the Hare/Niemeyer method (Bundeskanzlei, 2025, p.18). Of the nine cantons analyzed here, eight previously used Hagenbach-Bischoof for at least the majority of their parliamentary seats, only Graubünden priorly used majority voting throughout. Zug's Neuheim constituency and Uri's smaller municipalities also used Majorz rather than Hagenbach-Bischoff (Bundeskanzlei, 2013, Table 1; BGE 136 I 376 E. 5.1 ; Standeskanzlei Uri, 2020, p. 1; Statistisches Amt des Kantons Zürich, 2011, p. 286; Regierungsrat des Kantons Schaffhausen, 2007, p. 6; Regierungsrat des Kantons Aargau, 2007, p. 18).

4.1.2 The Hagenbach-Bischoff Method

To be able to understand later the differences between the HB and the DP Method one has to understand the allocation under the priorly used Hagenbach-Bischoff system.

Under Hagenbach-Bischoff, seats are allocated separately in each constituency. Let M denote the number of seats in the constituency, v_i the votes for an eligible list i, and V the total list votes used to calculate the distribution number. The distribution number D is the smallest integer strictly greater than $V/(M+1)$. In the first allocation, each list receives the integer part of its votes divided by D (Bundeskanzlei, 2025, p. 41):

$$D = \left\lfloor \frac{V}{M+1} \right\rfloor + 1, \quad s_i = \left\lfloor \frac{v_i}{D} \right\rfloor.$$

If the first allocation leaves seats unassigned, the remaining seats are awarded one at a time. For each list, its votes are divided by the number of seats it currently holds plus one. The list with the highest quotient receives the next seat; its quotient h_i and the allocated seats s_i are then recalculated, and the process continues until all M seats have been filled:

$$h_i = \frac{v_i}{s_i + 1}.$$

This calculation yields the same allocation as the D'Hondt divisor method with downward rounding (Bundeskanzlei, 2025, p. 41). List alliances, eligibility thresholds, and rules for

resolving equal quotients depend on the applicable electoral law. The counterfactual in this thesis applies the relevant former cantonal rules within each constituency, as detailed in the Data Cleaning section.

The central weakness of the traditional cantonal application of Hagenbach–Bischoff does not arise solely from the allocation formula, but from applying it separately in constituencies that often have few seats. Small district magnitudes create high and geographically varying natural thresholds, meaning that votes for smaller parties may fail to contribute to a seat and cannot be pooled across the canton. This can impair equality of electoral success and systematically disadvantage smaller parties. As shown in the later section on the adoption of Double Pukelsheim, these inequalities were central to the legal and political debates that led several cantons to adopt Double Pukelsheim. (Pukelsheim & Schuhmacher, 2004, pp. 507-508, 521; BGE 129 I 185 E. 7.1.1 and E. 7.6.2)

4.1.3 The Double Pukelsheimer Method

Based on the weaknesses of the Hagenbach-Bischoff Method and the FSC decision, the canton of Zurich comissioned the mathematician Friedrich Pukelsheim to develop a new method (Lüscher, 2024, p.10). The following section explores how the algorithm works and how it makes sure that it complies with the Federal Law and mitigates the HB weaknesses.

Based on Article 8 and 34 of the Federal Constitution, there are three principles which have to be followed. These are principle of equal voting value, equality of voting power and equality of electoral value. The equal voting value ensures that each voter within an electoral district gets allocated the identical number of votes. The second principle is the equality of voting power (voting weight guarantee). This requires the population represented per seat to be approximately equal across electoral districts. Thirdly there is the equality of electoral success, which should assure, that all votes contribute equally to the election and that as many votes as possible are taken into account in the allocation of seats. (Pukelsheim & Schuhmacher, 2004, pp. 506-507) Districts with fewer than nine seats have a natural quorum above 10% under the approximation $1/(M+1)$, where M is the number of seats. The Federal Supreme Court treats 10% as a target for natural quorums when redesigning an electoral system. Exceptionally, compelling historical or regional minority-protection reasons may justify a higher natural quorum. (Pukelsheim & Schuhmacher, 2004, p. 505; BGE 131 I 74 E. 5.2–5.4) Equality of counting value concerns the formal treatment of votes, while equality of voting power depends on the separately determined ratio of population to seats in each electoral district (Pukelsheim & Schuhmacher, 2004, pp. 507, 515–516). For the traditional application of Hagenbach–Bischoff, the main difficulty lies in equality of electoral success when small districts impose high natural thresholds (Pukelsheim & Schuhmacher, 2004, p. 508; BGE 129 I 185 E. 7.1.1 and E. 7.6.2).

To approach this main issue of the Hagenbach-Bischoff, the Double Pukelsheimer or originally called the “Neues Zürcher Zuteilungsverfahren” knows a upper and a lower apportionment (Pukelsheim & Schuhmacher, 2004, p.513).

- **Upper Apportionment** To bypass the natural quorums, in the upper apportionment, the whole canton is seen as one single district. All the seats are then allocated towards the parties. For that the votes from the district-level have to be summarized on a cantonal level, there the first difficulty emerges. Since the district magnitudes vary, the voters also have a different amount of votes, depending on from which canton they are from. Therefore to

be able to summarize on a global level, the following is done. The number of party votes per constituency are taken and divided with the respective number of seats. The resulting quotient gets rounded to the next number and indicates the amount of voters per list. The rounded quotients, can then be summed across all districts and divided by the cantonal divisor to get each parties total seats. The divisor is set by an official authority, in Zurich e.g. its the Department of the Government Councillor which is responsible for the elections. The department needs to assure, that with the set divisor exactly as many seats can be allocated as there are available. The number of seats which result per party, are rounded again to the next number. (Pukelsheim & Schuchmacher, 2011, p.1584-1585 ; Kanton Zürich, 2003, § 10 and 103)

- **Lower Apportionment** This step assigns each party's cantonwide seats from the upper apportionment to its district lists while preserving each district's predetermined seat total. First, the party votes in each district are divided by a district divisor chosen so that the rounded results add up to the district's seats. These initial allocations may not match the parties' cantonwide entitlements: in the 2011 Zurich election, the SVP would have received 55 seats instead of the 54 assigned in the upper apportionment. To correct such differences, every district list of a party is divided by the same additional list group divisor. A divisor above one reduces the party's unrounded quotients and can remove seats while a divisor below one can add seats. Consider a hypothetical SVP list in District Y whose quotient after division by the district divisor is 2.51, giving it three seats after rounding. If increasing the SVP's list group divisor to 1.01 changes no other SVP list's rounded allocation, this quotient falls to $2.51/1.01 \approx 2.49$, so the list receives two seats and the SVP's total seats match the ones from the upper apportionment. This change though may leave District Y one seat short, requiring an adjustment to its district divisor. The party and district divisors are then adjusted in turn until every party receives its cantonwide entitlement and every district retains its fixed number of seats. (Pukelsheim & Schuhmacher, 2011, pp. 1585–1586)

To summarize, the upper apportionment in the Double Pukelsheim system reduces inequalities in electoral success caused by the high natural thresholds of small constituencies under the previous district-by-district Hagenbach-Bischoff system. It does so by pooling parties votes across the canton before seats are assigned to constituencies. This is also the main improvement of DP regarding the three principles in contrast to Hagenbach-Bischoff.

4.1.3.1 Trade-offs and Criticisms

Alike every proportional system, also the Double Pukelsheimer system has its flaws for which it gets criticized. One of the main weaknesses of the biproprtional apportionment compared to the HB System is its complexity. It gets mentioned throughout, that its relatively difficult to understand the system holistically and that there is profound knowledge needed, to be able to compute the seat allocation. (Bundeskanzlei, 2013, p. 19 ; Bundeskanzlei, 2025, p.24 ; Lüscher, 2024, p.20) Lüscher tough mentions, that the usage of list alliances under the Hagenbach-Bischoff method also led to a high complexity situation. This especially since, beforehand the election it was impossible to estimate, which party would profit from the alliance. (Lüscher, 2024, p.20) Another main point of criticism, are the so called "discordant seat allocations". The discordant seat allocations describe the phenomena when within a district a party wins more seats than another party with more votes in that district. This phenomena can appear through the lower apportionment process and the dual role of the system to achieve both proportion-

ality canton wide and within a district. The emerging situation, can cause confusement and frustration especially within the party which initially received more votes in that specific constituency. (Pukelsheim & Schuhmacher, 2004, p. 519) Not only for the parties themself it can be irritating, but also for the voters. This especially since in Switzerland elections often contain a personal vote component, if then a popular figure from one party, loses its seat to a less known figure from another party and district, this may raise concerns. A reaction to this flaw of the DP-System, was that the cantons of Zug, Schaffhausen and Graubünden implemented a majority clause, which ensures that the most popular party in the corresponding constituency receives atleast one seat. (Lüscher, 2024, p. 21) Pukelsheimer, himself was already aware of this issue in 2004 and pointed out that the so called majority requirement can in rare ocassions be not fulfilled. On base of this he mentioned the possibility of the implementation of such a majority clause. (Pukelsheim & Schuhmacher, 2004, p. 520) Another issue of the system, is that different turnout level and different proportions of residents who are eligible to vote in the consituencies can determine the allocated mandates of other constituencies. To illustrate this , Lüscher provides an example: There are three Districts with the same size and number of mandates, but while District 1 produces 50% of all votes, District 2 and 3 only produce each 25%. Through this District 1 determines 50% of the canton-wide party result, while only having 33.3% of the mandates. Subsequently this leads to under- and overrepresentation in the corresponding districts. (Lüscher, 2023, pp. 578-579) One other point was often discussed to be a possible weaknes of the Double Pukelsheim: Party System Fragmentation. For example both reports of the Swiss Federal Chancellery in 2013 and 2025 mention the possible party fragmentation in parliaments. As a possible measure to prevent fragmentation, the implentation of thresholds is mentioned. (Bundeskanzlei, 2013, p.18 ; Bundeskanzlei, 2025, p. 24) Though Lüscher, came in his Thesis to the perception, that the general trend of a rising amount of parties in swiss cantonal parliaments, is unrelated to the implementation of the Double Pukelsheim. He then also mentioned, that the inexistency of an effect, might come from that several cantons have implemented thresholds. (Lüscher, 2024, p. 135) Therefore it can't be conclusively stated, whether the reform would have had an impact on party system fragmentation without the implentation of thresholds or not.

Taken together, these criticisms illustrate the central trade-off of Double Pukelsheim. Although the system improves canton-wide proportionality and reduces the number of ineffective votes, it does so at the cost of greater computational complexity and potentially counterintuitive outcomes at the constituency level. Majority clauses and electoral thresholds can mitigate discordant seat allocations and party-system fragmentation, but may simultaneously weaken proportionality. The system therefore does not eliminate electoral distortions entirely, it rather prioritizes party proportionality and equal vote success over simplicity and fully autonomous constituency-level representation.

4.1.4 The Adoption of Double Pukelsheim in Swiss Cantons

In the following paragraphs the arguments brought up in each reform canton will be discussed in chronological order. The year in the beginning of each entry, marks the year of first application of the double Pukelsheimer in the corresponding canton.

- **2007:** Zurich became the first canton to use Double Pukelsheim, which it had commissioned Friedrich Pukelsheim to develop. The reform followed a 2002 voting-rights complaint about elections to Zurich's city parliament. The Swiss Federal Supreme Court partially upheld

the complaint. It found that the thresholds in some districts with up to 33% were massively above the 10% threshold, which is seen as compatible with the prinicples of proportional representation. Further was it claimed, that thresholds vary to much between districts. The ruling required the city of Zurich to reform its electoral system; the canton subsequently implemented the Pukelsheimer Proporz for cantonal elections with a 5% district-level threshold for parties to enter parliament (BGE 129 I 185 E. 7.1.2, E. 7.6.2–7.6.3 and E. 8.3.2; Lüscher, 2024, pp. 9–10, 33-36). In 2021, Zurich added a 3% cantonwide threshold, first applied in the 2023 election. A change to how that threshold is calculated was adopted in 2025 and will first apply in the 2027 election (Regierungsrat des Kantons Zürich, 2024, pp. 2, 5; Regierungsrat des Kantons Zürich, 2025, p. 1).

- **2008:** Schaffhausen: In a 2004 popular vote, Schaffhausen's electorate approved a reduction of the cantonal parliament from 80 to 60 seats. Shortly after a motion calling to reform the electoral law to better protect smaller parties was submitted. After this the Schaffhausen Government proposed a reform of the electoral system and the implementation of the biproportional apportionment. The biproportional apportionment was also preferred, because of the Supreme Courts decisions in Zurich and Aargau. (Regierungsrat des Kantons Schaffhausen , 2007, pp. 1, 3, 13-14) Subsequently the parliament (2007) and the voters (2008) approved the changes. In 2015, the cantonal parliament added a majority clause, which ensures that the strongest party receives at least one seat in the corresponding district. (Lüscher , 2024 , pp. 42- 43) Further did the canton of Schaffhausen decide to prohibit list alliances (Kanton Schaffhausen, 1904/2019, Art. 2b para. 3).
- **2009:** Aargau's voters in the year 2003 approved a proposal in a popular vote, which wanted to lower the amount of parliament seats from 200 to 140. This decision naturally raised the thresholds in the districts and therefore a second constitutional amendment allowing to merge small districts was proposed and approved by the voters. Though the parliamentarians decided against making use of it. As a consequence of the parliaments decision, the electoral equity wasn't given anymore. The Government of Aargau had to act and proposed to the Grand Council to either merge districts or to change to the Double-Pukelsheim. The Grand Council again decided to not change anything. Shortly after this decision by the Grand Council the FSC issued in a verdict, that there are existing inequalities, because of the varying thresholds and district sizes. The court ordered that this needs to be changed. In September 2007, the Grand Council with a narrow majority then decided to implement the Double-Pukelsheimer. In 2011 a 5% local (district) and a 3% global quorum was added after approval of the Grand Council and the voters (Lüscher, 2024, pp. 38 - 40)
- **2014:** Nidwalden: In the year 2010 members of the Green Party in Nidwalden filed a complaint regarding the seat allocation process for the cantonal elections (Lüscher, 2024, p. 44). That complaint got rejected initially by the Cantonal Court but later the cantonal decision got overruled by the FSC. The Court rejected the argument that historical practice justified the allocation method (BGE 136 I 352 E. 4.; Lüscher, 2024, pp. 44–45). As a reaction to this decision the parliament of Nidwalden came up with multiple possible reforms. The most popular of the suggested reforms, was the change towards the biproportional system. The parliament and the electorate subsequently approved the implementation of the Double-Pukelsheimer. (Lüscher, 2024, p.45)

- **2014:** Zug: In 2010, representatives of several parties formed a complaint at the FSC regarding the differently sized districts. The Federal Supreme Court agreed on the appeals main point (BGE 136 I 376 E. 4.5). As a reaction to this decision the cantonal Government presented three different possibilities on how to adapt to the FSC ruling. The responsible comission favoured one option: adopting the Double Pukelsheim. The Parliament adopted the reform and added based on the model of Aargau a 5% local and 3% global threshold. (Lüscher, 2024, p. 47-51) A competing proposal to retain the existing system was struck down by the FSC in 2013 (BGE 139 I 195 E. 4). In addition to the thresholds, the canton also implented a majority clause, ensuring that the most popular list within a district always receives at least one seat (Kanton Zug, 2006/2026, § 52e paras. 1a–1b). Voters approved the biproportional reform on 22 September 2013 (Lüscher, 2024, p. 47-51).
- **2016:** Schwyz: The reform process began with the approval of the new cantonal constitution by the electorate. In that it was foreseen, to keep the current electoral system, which means that each municipality is its own electoral district. (Lüscher, 2024, p. 53) In September 2011, the Cantonal Government prepared the Cantonal Council Election for 2012, against the decisions made there three individual complaints were raised at the FSC. The main complaint was that the thresholds in some of constituencies were too high. The Court agreed on the complaints of the three parties partly and stated that the current electoral system doesn't meet the requirements of a proportional representation system (Urteil des Bundesgerichts 1C_407/2011 etc. vom 19. März 2012, Sachverhalt A–B und E. 5.3, 6). Later also the Federal Assembly refused to approve the critical sections in the new constitution of Schwyz. As a reaction towards this two different initiatives were formed by the political parties in the canton, one wanted a majority system and the other wanted fairer proportional representation with the willingness to remove guaranteed seats for municipialities. The government reacted with the proposal to inaugurate the DP-System while keeping municipialites as districts and preserving their guaranteed seats. The parliament supported putting the counterproposal together with a 1% global threshold to voters. The voters could then decide on either the majority vote initative or the counterproposal. The counterproposal and with that the implementation of the DP was approved. (Lüscher, 2024, pp. 53-55)
- **2017:** Valais: In the year 2004, a complaint at the FSC was filed regarding the varying thresholds in the electoral districts. The FSC then rejected the complaint and argued that there are historical reasons for the shapes of the electoral districts. (BGE 131 I 85 E. 2.3–2.5) Six years later an alliance from Upper-Valais proposed the popular initiative "Every Vote Counts" and with that the implementation of the DP with only three new districts. In 2011 the canton then set up a commission to develop proposals. (Lüscher, 2024, pp. 58–59) In 2012 multiple voters again filed a complaint at the Supreme Court it then changed its position and ruled that the existing system violated the Federal Constitution (BGE 140 I 107 E. 4–5). The parliament then approved a constitutional reform introducing biproportional seat allocation, but it also added a guarantee of 35 seats for German-speaking Upper Valais. Voters narrowly rejected the package and also the initiative "Every Vote Counts" in 2015. To ensure lawful elections in 2017, parliament unanimously adopted a temporary decree in March 2016 without the seat guarantee. The resulting system retained 14 subdistricts, grouped them into six larger districts, and applied biproportional allocation separately within each larger district. Parliament incorporated these rules into the cantonal electoral

law in December 2017. (Lüscher, 2024, pp. 61–63) The Canton Valais also implemented a 8 % local (district-level) threshold (Kanton Wallis, 2004b, Art. 154 Abs. 1).

- **2020:** Uri used a mixed electoral system: municipalities with one or two seats elected their representatives by Majorz, whereas those with at least three seats used proportional representation. A complaint against the seat allocation for the 2016 election led the Federal Supreme Court to rule that the allocation method within the Proporz municipalities violated electoral equality. The Court did not reject the mixed system itself, which it considered justifiable because it preserved personality-oriented elections and an autonomously elected representative for very small municipalities (BGE 143 I 92 Sachverhalt, E. 5–5.4 and E. 6.5 ; Kanton Uri, 2019, pp. 6–7). The government subsequently proposed Double Pukelsheim for all eight municipalities then using proportional representation. During the legislative debate, however, the Landrat raised the threshold for Proporz from three to five seats, moving Attinghausen, Flüelen, Seedorf, and Silenen to Majorz and leaving only Altdorf, Bürglen, Erstfeld, and Schattdorf in the Double Pukelsheim allocation. The parliamentary majority reasoned that Majorz better suited the four municipalities' small populations and strong autonomy. It also pointed to the weak role or absence of local party organizations, occasional uncontested elections, and electoral statistics suggesting that voters chose personalities rather than party lists. The government opposed the extension and, based on a constitutional-law opinion, warned of a high risk that it would fail a renewed challenge before the Federal Supreme Court. Uri's voters approved both the constitutional amendment extending Majorz and the separate Proporzgesetz reform in May 2019 (Kanton Uri, 2019, pp. 4, 10–11; Lüscher, 2024, pp. 64–67).
- **2022:** Graubünden: Before the Implementation of the biproportional apportionment the canton still used a majority based electoral system. Though there were multiple attempts to introduce a PR system. A main reason for these attempts to fail, was the fear that the protection of local minorities and the 20 single-seat-constituencies would be weakened. (Lüscher, 2024, p.69) In 2017 then multiple parties and individuals filed a complaint at the Federal Supreme Court after the cantons government had released the seat allocations for the districts in the upcoming election. The court partially upheld the complaint, it especially decided that the seat allocation for the smallest district (Avers) and the six largest districts needed to be revised. Specifically the Court found that the guaranteed seats for Aver's 160 eligible voters violated equality of voting weight, while majoirtiy voting in the six largest constituencies unjustifiably impaired the equal effect of votes on seat allocation. (BGE 145 I 259 E. 7.3 and E. 8.5.3) As a response to the FSC decision, the government launched a consultation process after which, it presented nine different models. While the government preferred the DP-System, the municipalities and large parties favoured a mixed electoral model. The willingness of the Socialdemocrats to withdraw a separate initative helped build support for Double-Pukelsheim. The parliament then chose the Double Pukelsheim as a compromise, since it made the seat allocation more proportional while keeping representation for small districts. A 3% global (cantonwide) threshold was also introduced. In addition to the thresholds, the canton added a majority clause similar to the one in Schaffhausen and Zug (Kanton Graubünden, 2021/2026, Art. 28 paras. 2–4). In 2021 the electorate voted in favor of implementing the DP-System. (Lüscher, 2024, pp. 71-72)
- **2027:** Basel-Landschaft: Unlike other cantons, in Basel-Landschaft there were no implicit quorums of more than 10%, which means that the law complied with the FSC ruling. In 2015

though a parliamentarian requested the cantonal government to examine the comparative effects of the current system with the Double Pukelsheimer. The Government didn't see any need for a reform, the parliamentarian though kept working on the topic and demanded the establishment of a special comission to review the topic. Although the parliament also rejected this proposal, it instructed its executive board to work with external experts to examine how the electoral system could better reflect party vote shares while preserving elected representatives' local ties. A few years later in 2022, the executive board came to the conclusion, that it recommended the implementation of the DP. Through the support of the parliament, the government was then instructed to draft a revision of the electoral law. (Lüscher, 2024, p.86) In the year 2024, the government then submitted the bill and the Cantonal Council voted in favour of it (Regierungsrat des Kantons Basel-Landschaft, 2024 ; Landrat des Kantons Basel-Landschaft, 2024, p. 1052). The new reform was subject to a mandatory referendum, the canton's voters voted for the implementation in 2025 (Kanton Basel-Landschaft, 2025b). The canton also decided to implement a 3% global and a 5% local threshold and stipulated that each constituency has one guaranteed seat (Kanton Basel-Landschaft, 1981/2025, Art. 39a and Art.49 lit.a). The first election under the new system will be held in 2027 (Kanton Basel-Landschaft, 2026).

To summarize, in 8 of 10 reform cantons, a raised complaint at the FSC led to a later reform. Only the cantons Schaffhausen and Basel-Landschaft reform process took place without a Federal Supreme Court ruling. Nevertheless the canton of Schaffhausen's decision was directly influenced by the FSC's decisions in Aargau and Zurich. Throughout all complaints, the concern was electoral equality. In seven of them excessively high natural thresholds in proportional constituencies were central. While in Graubünden the concerns were mainly raised by the use of of the majority system for its large districts and the allocation of seats among constituencies. This demonstrates the influence of the 2002 Federal Supreme Court decision regarding Zurich's City Parliament.

Beyond their different paths to reform, the cantons also implemented Double Pukelsheim under substantially different institutional conditions. The number of seats included in the Double Pukelsheim allocation varies from 37 in Uri where the system applies only to four municipalities to 180 in Zurich (Standeskanzlei Uri, 2020, p. 1; Kanton Zürich, 2003, § 88). Moreover, some cantons introduced explicit local or canton-wide electoral quorums, whereas others did not, potentially limiting smaller parties' ability to benefit from the more proportional allocation method. These arrangements have also changed over time—for example through the later introduction of thresholds or majority clauses—while party competition and vote distributions vary between elections. This institutional variation motivates the exploratory comparisons below: whether smaller parties' seat-share gains differ by the number of seats allocated through Double Pukelsheim, by the presence of an explicit quorum, and across successive elections within the same canton.

4.2 Literature Review

Building on the institutional mechanisms described above, the existing empirical literature generally finds that Double Pukelsheim improves vote–seat proportionality and benefits parties that were disadvantaged by small electoral districts. However, the studies differ substantially in their research designs, geographical coverage, and treatment of counterfactual electoral rules.

Pukelsheim and Schuhmacher provide an early counterfactual comparison based on the 2011 Zurich election. They apply the former Hagenbach–Bischoff method to the observed votes while assuming four hypothetical list alliances based on political orientations and historical alliances. Under this counterfactual, the larger SVP, SP, and FDP receive additional seats, primarily at the expense of smaller parties such as the BDP and Alternative List. Their calculation also indicates considerably greater inequality in the electoral success of votes under Hagenbach–Bischoff. The analysis nevertheless concerns only one election, and its results depend partly on assumptions about which list alliances parties would have formed. (Pukelsheim & Schuhmacher, 2011, pp. 1590–1592)

Lüscher conducts a similar comparison for the 2016 Schwyz election. Under his Hagenbach–Bischoff counterfactual, which does not assume any list alliances, the dominant SVP and CVP would have benefited, whereas the left alliance and especially the GLP benefited from Double Pukelsheim. Most notably, the GLP's three observed seats would have been reduced to zero under Hagenbach–Bischoff (Lüscher, 2024, p. 13). This example illustrates the potentially substantial effect of the allocation rule on an individual smaller party, but it does not establish whether effects of this magnitude occur systematically across cantons and elections.

Leuzinger broadens the comparison by examining the most recent election in each of the seven cantons that had introduced Double Pukelsheim by 2018. His study is the closest precedent for the present analysis because it calculates hypothetical seat distributions under the former cantonal electoral systems. However, it covers only one election per canton and assumes that the list alliances observed in the final pre-reform election would have persisted (Leuzinger, 2018, p. 42). The present thesis does not carry these historical alliances forward. Consequently, the two counterfactual specifications can produce different Hagenbach–Bischoff seat allocations. This illustrates the sensitivity of counterfactual results to assumptions about unobserved alliance formation. Leuzinger also mentions that he observed, that mainly smaller parties profited from the transition and that around 10% of the mandates mechanically change the party, through the transition (Leuzinger , 2018, p.42).

Taken together, the counterfactual calculations by Pukelsheim and Schuhmacher, Lüscher, and Leuzinger constitute the closest methodological precedents for the present thesis. They apply a former allocation rule to votes observed under Double Pukelsheim and compare the resulting counterfactual seat distribution with the actual allocation. Their comparisons therefore hold the observed post-reform vote distribution constant and isolate the difference produced by the allocation rules.

The interpretation of this difference can be clarified using Fiva and Folke's framework for separating the mechanical and psychological effects of electoral reform. Fiva and Folke examine a reform of Norwegian municipal elections, first applied in 2003, that replaced D'Hondt with modified Sainte-Laguë. (Fiva & Folke, 2016 , p.266) Their principal estimate of the mechanical reform effect compares the actual pre-reform allocation with a counterfactual application of the new formula to the same pre-reform votes. They also apply the former and new formulas to the same post-reform votes to examine how behavioral responses may alter the magnitude of the mechanical effect. (Fiva & Folke, 2016, pp. 270–272) The Swiss studies described above, and the present thesis, follow this latter computational approach: the former formula is applied counterfactually to votes observed under the new system. This measures the mechanical allocation difference conditional on the observed post-reform vote

distribution, but it does not estimate how parties and voters would have behaved had the former system remained in place or identify the reform's complete causal effect.

Rather than recalculating individual elections, Lüscher's principal empirical analysis estimates how cantonal political outcomes would have developed without the reforms. He finds that Double Pukelsheim substantially reduced vote–seat disproportionality, with stronger improvements in cantons that previously had smaller average district magnitudes (Lüscher, 2024, pp. 129–143). However, Lüscher finds no significant increase in the effective number of parliamentary parties or electoral candidacies. Voter turnout initially declined slightly, but this effect diminished over subsequent elections. These findings distinguish the reform's immediate mechanical consequences from its more limited behavioural effects on parties and voters.(Lüscher, 2024, pp. 143–145)

The existing literature therefore provides strong evidence that Double Pukelsheim improves proportionality, as well as several examples in which smaller parties gain seats. What remains less systematically examined is the magnitude and consistency of the conditional mechanical allocation difference across successive elections. Existing counterfactual studies either focus on individual elections or cover only one election per reform canton, while Lüscher's longitudinal analysis examines aggregate disproportionality and broader political outcomes rather than the election-specific seat gains of smaller parties.

This thesis addresses that narrower gap by applying a consistent Hagenbach–Bischoff counterfactual to 30 elections in eight transition cantons. Two elections from Graubünden are additionally presented as contextual comparisons. Because Graubünden previously used majority voting rather than Hagenbach–Bischoff, these allocations are hypothetical and are excluded from the main transition estimate. The thesis holds observed post-reform votes constant, applies a common definition of smaller parties, and measures their change in seat share for every election. In contrast to Leuzinger (2018) and Pukelsheim & Schuchmacher (2011), it does not assume that hypothetical or historical pre-reform alliances would have persisted. The analysis therefore estimates the mechanical allocation difference conditional on the observed votes without introducing an additional behavioral assumption about counterfactual alliance formation.

4.3 Expected Effects of the Transition

Based on the theoretical considerations developed in the preceding sections, several effects can be expected from the transition from the Hagenbach–Bischoff method to the Double Pukelsheim system. Under a district-by-district Hagenbach–Bischoff allocation, small district magnitudes create high natural thresholds for winning a parliamentary seat. Votes cast for parties that fail to obtain a seat in a district cannot contribute to their representation elsewhere. This produces ineffective votes, disproportionately disadvantages smaller parties, and may advantage larger parties. Double Pukelsheim instead pools votes across the whole area covered by the upper apportionment before distributing each party's seats among the districts. This should increase the equality of electoral success and reduce vote–seat disproportionality. Mechanically, and conditional on the same observed votes, smaller parties that surpass the applicable legal thresholds are therefore expected to receive a larger seat share under Double Pukelsheim, while the seat advantage previously enjoyed by larger parties should decline (Pukelsheim & Schuhmacher, 2004, pp. 506–508, 513-514 , 518).

The magnitude of this mechanical difference should depend particularly on district magnitude under the Hagenbach–Bischoff comparison. The fewer seats allocated in a district, the larger the vote share a party generally needs to obtain representation. Double Pukelsheim should therefore produce larger gains for smaller parties where the former or counterfactual Hagenbach–Bischoff allocation consists of many low-magnitude districts than where seats are already allocated in relatively large districts. This expectation is consistent with Lüscher's finding that the reduction in disproportionality was stronger in cantons with smaller pre-reform average district magnitudes (Lüscher, 2024, pp. 132–133).

The effect is also expected to depend on the specific institutional design adopted by each canton. Reflecting Switzerland's federal structure, the reformed cantons differ in their use of canton-wide and district-level legal thresholds ranging from 1 to 8 percent, majority clauses, and other supplementary provisions. Explicit electoral thresholds can restrict gains among smaller parties by excluding lists that do not reach the required vote share and can therefore partly offset the greater proportionality produced by the upper apportionment (Bundeskanzlei, 2013, p. 18; Lüscher, 2024, pp. 107, 135). The expected effect of majority clauses is less unambiguous because they protect the strongest list within a district rather than parties of a particular size; they may modify individual district allocations without systematically benefiting either smaller or larger parties (Pukelsheim & Schuhmacher, 2004, p. 520).

Valais and Uri are particularly important special cases because Double Pukelsheim does not cover the canton through a single canton-wide upper apportionment. Valais conducts separate upper apportionments in six electoral regions, so votes aren't pooled across the entire canton. This is expected to limit the proportionality-enhancing effect relative to a fully canton-wide implementation, particularly for parties whose support is dispersed across regions (Lüscher, 2024, p. 57). In Uri, Double Pukelsheim applies only to Altdorf, Bürglen, Erstfeld, and Schattdorf, while the remaining municipalities use Majorz. This limits the territorial and numerical scope of the reform, although it does not necessarily reduce the mechanical effect among the four municipalities included in the upper apportionment (Kanton Uri, 2019, pp. 4, 10–11; Lüscher, 2024, pp. 64–67).

The total number of seats allocated through Double Pukelsheim may provide further context. All else equal, a larger number of seats allows vote shares to be translated into seat shares more finely (Lüscher, 2024, p. 106). However, parliament size does not generate an unambiguous expectation for the difference between the two allocation methods because the Double Pukelsheim and Hagenbach–Bischoff allocations being compared distribute the same total number of seats. How those seats are divided among districts under Hagenbach–Bischoff is the more direct mechanism. Differences between larger and smaller parliaments should therefore be interpreted as exploratory contextual evidence rather than as an isolated effect of parliament size.

The estimated difference may additionally depend on the electoral rules used to construct the counterfactual outcomes, including the treatment of list alliances and canton-specific allocation provisions. It represents the mechanical allocation difference conditional on the observed post-reform votes, not the reform's complete causal effect on party or voter behaviour. No uniform directional development across successive elections is expected because party vote shares, the parties classified as smaller, district magnitudes, thresholds, and other electoral rules can change over time.

Together, these expectations provide the foundation for the following hypothesis and exploratory questions concerning differences in smaller parties' seat shares.

5 Hypotheses

To answer the research question, the thesis tests one main hypothesis concerning how the transition from the Hagenbach–Bischoff system to Double Pukelsheim mechanically affects the seat shares of smaller parties. Complementary exploratory analyses evaluate whether this effect varies with district magnitude, the presence of an explicit electoral threshold, and successive elections since implementation within each canton. Together, these analyses identify both the immediate and potentially evolving distributional consequences of the electoral-system reform.

5.1 Hypothesis 1

For each election e, smaller parties or electoral lists are those with less than 10% of the vote in the *analyzed area*: the whole canton except in Uri, where the *analyzed area* comprises only the four municipalities using Double Pukelsheim. The 10% threshold draws on the Swiss Federal Supreme Court's distinction between a 10% upper limit for direct quorums and a 10% target for natural quorums when an electoral system is redesigned (BGE 131 I 74 E. 5.3–5.4; Bundeskanzlei, 2013, p. 8). The legal benchmark therefore provides a substantively relevant cutoff.

Let S_e^{DP} and S_e^{HB} be their combined seats under Double Pukelsheim and the counterfactual Hagenbach-Bischoff allocation, respectively, and let M_e be the number of seats allocated through Double Pukelsheim and included in the comparison. Their gain in seat share, shown for each election in Figure 2, is

$$\Delta_e = 100\frac{S_e^{DP} - S_e^{HB}}{M_e} \quad \text{percentage points} .$$

The hypothesis uses the unweighted mean, $\bar{\Delta} = \frac{1}{E}\sum_{e=1}^{E} \Delta_e$, so each election counts equally. The analysis also reports a mean weighted by the number of analyzed seats in each election.

$\boldsymbol{H_1}$**:** Across the analyzed elections, the mean mechanical difference ({}) is positive: electoral lists or list groups with less than 10% of the vote receive, on average, a larger share of the analyzed seats under Double Pukelsheim than under the counterfactual Hagenbach–Bischoff allocation.

5.2 Exploratory Questions

5.2.1 Exploratory question 1: Differences by district magnitude

Does the gain in smaller parties' seat share under Double Pukelsheim relative to the counterfactual Hagenbach–Bischoff allocation tend to be larger in cantons with smaller electoral districts? District size is measured primarily by the mean district magnitude in the Hagenbach–Bischoff comparison. The median district magnitude is also reported to capture the size of the typical district and indicate whether the mean is influenced by a small number of unusually large or small districts. For election e, with D_e districts and district magnitudes m_{de}, the mean district magnitude is

$$\bar{m}_e = \frac{1}{D_e} \sum_{d=1}^{D_e} m_{de} = \frac{M_e}{D_e}.$$

A lower mean or median district magnitude is expected to be associated with a larger gain Δ_e for smaller parties. As a complementary measure, the analysis reports the share of districts with fewer than nine seats,

$$q_e = \frac{1}{D_e} \sum_{d=1}^{D_e} \mathbb{1}(m_{de} < 9).$$

The nine-seat cutoff follows the Swiss Federal Supreme Court's reasoning: under the natural-quorum approximation $1/(M+1)$, districts with fewer than nine seats impose a threshold above 10%. A larger value of q_e is therefore expected to be associated with a larger gain Δ_e (BGE 131 I 74 E. 5.3–5.4 ; Bundeskanzlei, 2013, p. 8). Because the number of transition cantons is small and their electoral rules differ, these relationships are interpreted descriptively rather than as isolated causal effects of district magnitude. Graubünden is shown only as a contextual comparison because its pre-reform system was Majorz and its Hagenbach–Bischoff allocation is hypothetical (Bundeskanzlei, 2013, Table 1).

5.2.2 Exploratory question 2: Differences by DP seat count (Parliament size)

How is the number of seats allocated through Double Pukelsheim associated with the average gain in smaller parties' seat share under Double Pukelsheim relative to the counterfactual Hagenbach–Bischoff allocation?

For each canton, the election-level gains Δ_e are averaged first so that cantons with more observed elections receive no extra weight. The resulting canton-level mean is then examined in relation to the number of seats allocated through Double Pukelsheim. Because both allocation methods distribute the same number of seats within each election and district magnitude is the more direct mechanism, no directional relationship is hypothesized. The association is interpreted descriptively rather than as an isolated causal effect of the number of Double Pukelsheim seats. Graubünden is shown separately because its pre-reform system was Majorz and its Hagenbach–Bischoff allocation is hypothetical (Bundeskanzlei, 2013, Table 1).

5.2.3 Exploratory question 3: Explicit electoral quorums

How does the gain in smaller parties' seat share under Double Pukelsheim relative to the counterfactual Hagenbach-Bischoff allocation vary between elections conducted with and without an explicit electoral quorum?

An explicit quorum is a legally specified minimum vote share that a party or list group must reach at the district, regional, or cantonal level to qualify for seat allocation. Among the nine cantons analyzed, six applied an explicit quorum in at least one observed election; excluding Graubünden, this applies to five of the eight cantons in the main comparison, as detailed in Section 4.1.4. Unlike the natural threshold created by limited district magnitude, it directly excludes parties that do not satisfy the legal requirement.

Because such quorums may prevent smaller parties from benefiting from the pooling of votes under Double Pukelsheim, the following directional expectation is formulated:

On average, the gain in smaller parties' seat share under Double Pukelsheim relative to the counterfactual Hagenbach-Bischoff allocation is smaller in elections with an explicit electoral quorum than in elections without one.

This comparison is interpreted descriptively because the level and geographical application of the quorums differ across cantons and elections and coincide with other institutional differences.

5.2.4 Exploratory question 4: Change across successive elections

How does the gain in smaller parties' seat share under Double Pukelsheim relative to the counterfactual Hagenbach-Bischoff allocation evolve across successive elections within each canton?

This analysis extends Hypothesis 1 by following the seat shares under both allocation rules and their difference Δ_e from each canton's first election under Double Pukelsheim onward. It assesses whether the overall difference identified in Hypothesis 1 is persistent across elections or driven by particular election-specific outcomes.

Stable positive values of Δ_e would indicate a persistent mechanical advantage for smaller parties, whereas substantial fluctuations or changes in sign would suggest that the effect depends strongly on the conditions of individual elections. The within-canton perspective also limits the influence of relatively stable institutional differences that complicate comparisons across cantons. No directional trend over time is hypothesized because vote shares, the composition of parties below the 10% threshold, district magnitudes, and electoral rules can change between elections.

6 Research Design and Analysis

6.1 Research Design and Counterfactual Strategy

This thesis uses a deterministic counterfactual reallocation design. The main analysis covers all 30 cantonal parliamentary elections held under Double Pukelsheim in Switzerland through 2026 in the eight cantons in which Hagenbach–Bischoff previously governed the relevant proportional allocation. The two elections from Graubünden are included only as contextual comparisons because the canton previously used a majority system. At the computational level, the observations are electoral lists within districts and elections, while the primary unit of comparison is the election.

Each election is evaluated under two allocation regimes. The observed outcome is the official allocation under Double Pukelsheim, obtained from the relevant cantonal election authority. The counterfactual outcome is obtained by applying the specified former district-level allocation rules to the same election's observed votes, electoral units, seat magnitudes, and reported list structure. Each election therefore serves as its own comparison. The design does not rely on an untreated control canton or on pre- and post-reform trends.

The counterfactual is constructed in five steps.

First, the official election files are imported and standardized to obtain, for every election, electoral unit, and reported list or list group, the observed votes, seat magnitude, and official Double Pukelsheim seats. Party and list labels are harmonized across source files, and voter numbers or vote shares are reconstructed where required by the structure of the official data.

Second, the counterfactual inputs are defined by holding the observed votes, electoral units, seat magnitudes, and reported list configuration constant. Joint lists reported as one unit in the official results remain pooled, but historical or hypothetical alliances between separately reported lists are not introduced.

Third, the specified former allocation regime is applied separately within each electoral unit. Most counterfactual allocations use the district-level Hagenbach–Bischoff method described above. Where the former cantonal rules contained additional eligibility, quota, majority, or tie-breaking provisions, these are incorporated into the calculation. The applicable canton-specific rules and deviations are documented in the following subsection.

Fourth, the counterfactual seats obtained in the individual districts or municipalities are aggregated to the party-election level and merged with the official Double Pukelsheim seat totals.

Finally, the resulting datasets are validated by checking party coverage, vote and seat consistency, electoral-unit seat totals, and the total number of seats allocated under both regimes. Where the source data provide multiple measures of the official Double Pukelsheim result—such as upper-apportionment totals, lower-apportionment totals, reported list mandates, or elected-candidate counts—these measures are additionally reconciled. Exact ties are resolved using the applicable canton-specific rule where this can be implemented from the available data. If more than one legally possible allocation remains, the code either retains the alternative outcomes or stops the calculation. It therefore never assigns a seat according to the arbitrary ordering of rows in the dataset. The unresolved alternatives observed in Graubünden 2022 and Schwyz 2024 concern only parties above the 10% threshold and therefore do not affect the main smaller-party comparison.

The resulting contrast is interpreted as a mechanical allocation difference conditional on the observed post-reform votes, rather than as the reform's complete causal effect. Because these votes were cast under Double Pukelsheim, the analysis cannot determine how voters, parties, candidate supply, campaign strategies, or list alliances would have differed if the former electoral regime had remained in force.

6.2 Counterfactual Assumptions and Scope

Applying the general counterfactual strategy requires several canton-specific decisions concerning electoral rules, list alliances, electoral units, and seat magnitudes. This subsection explains these decisions and defines the scope of the comparison.

Across the nine analyzed reform cantons, the counterfactual applies a district-level allocation to the electoral units retained from each observed election. In eight cantons, this generally reconstructs the formerly applicable Hagenbach–Bischoff regime, subject to the exceptions summarized in Table 1.[2] In Graubünden, which previously used Majorz, the Hagenbach–

[2] Aargau (BGE 91 I 266, Sachverhalt A ; Staatskanzlei des Kantons Aargau, 2004, section "Neu: 140 statt 200 Mitglieder"); Graubünden (Kanton Graubünden, 2003/2018, Art. 27; hypothetical HB specification based on BGE 91 I 266, Sachverhalt A); Nidwalden (BGE 136 I 352, E. 3.2); Schaffhausen (Regierungsrat des Kantons Schaffhausen, 1979, §§ 2, 36–37; Regierungsrat des Kantons Schaffhausen, 2007, pp. 2, 6); Schwyz (Urteil des Bundesgerichts 1C_407/2011 vom 19. März 2012, E. 5.2); Uri (Kanton Uri, 2018, Art. 22–23, "Bisher"; Kanton Uri, 2019, pp. 4, 11); Valais (BGE 140 I 107, Sachverhalt, pp. 108–109; Kanton Wallis, 2004a, Arts. 154–156); Zug (Kanton Zug, 2006, §§ 46–47, 55–56; BGE 136 I 376 E. 4.2, 4.4 and 5.1); and Zurich (Bentz, 2003, p. 2; Kanton Zürich, 1983/2005, §§ 87–88b)

Bischoff allocation is entirely hypothetical and is therefore excluded from the main transition estimate. In every case, the observed votes, electoral-unit seat magnitudes, and reported list structure are held constant. The common Hagenbach–Bischoff procedure is described in the institutional background; this subsection therefore focuses on canton-specific adaptations and limitations of the reconstruction.

Importantly, the comparison concerns the complete election-specific allocation regimes rather than only the mathematical difference between the two apportionment algorithms. The observed outcome incorporates the Double-Pukelsheim rules applicable in the respective election, including any explicit electoral quorum or majority clause. Further the counterfactual applies the relevant former district-level rules to the observed votes and electoral structure. Consequently, differences in seat allocation may reflect both the change in apportionment method and associated differences between the applicable rule packages.

Table 1: Former electoral systems and counterfactual treatment by canton.

Canton	Pre-reform allocation	Counterfactual treatment	Canton-specific counterfactual rule
AG	HB in 11 electoral districts	Former HB allocation reconstructed	Standard HB procedure
GR	Majorz	Hypothetical district-level HB allocation in 39 electoral districts ; excluded from the main estimate	Hypothetical standard HB specification based on the former Aargau procedure
NW	HB in 11 municipalities	Former HB allocation reconstructed	Standard HB procedure
SH	HB in 5 multi-seat districts, majority rule in 1 single-seat district	Former mixed district-level system reconstructed	Ceiling-based distribution number; relative majority in the single-seat district; statutory tie-breaking where required
SZ	HB in 30 municipalities	Former HB allocation reconstructed	Standard HB procedure; unresolved exact ties retained as alternatives
UR	HB in 8 municipalities, Majorz elsewhere	HB reconstructed for the 4 municipalities using DP	Tie-breaking by initial remainder and then party votes; unresolved alternatives retained
VS	HB in 14 districts	Former district-level HB regime reconstructed	8% district eligibility threshold: ineligible lists' votes remain in the distribution-number calculation
ZG	HB in 10 municipalities; Majorz in Neuheim	HB applied in all 11 municipalities; hypothetical for Neuheim	Standard HB procedure; hypothetical HB application in Neuheim
ZH	HB in 18 electoral districts	Former HB allocation reconstructed	Tie-breaking by party votes, followed by lot; unresolved lots retained as alternatives

Although all canton-specific provisions in Table 1 several do not affect the smaller-party outcomes. The statutory tie-breaking rules in Schaffhausen, Uri, and Zurich were not triggered in any analyzed election. In Schaffhausen, the statutory distribution number was 5,022 rather than 5,023 in Klettgau in 2008 and 2,196 rather than 2,197 in Neuhausen in 2024. These differences changed neither the initial nor the final seat allocation. In Schwyz 2024 and in Graubünden 2022, an unresolved exact tie produced alternative HB allocations for parties above the 10% threshold. Thus it only affected individual party totals but not the aggregated seat total of smaller parties used in the main analysis. The relative-majority allocation in

Schaffhausen's single-seat district was nevertheless applied and remains part of the reconstructed former system.[3]

Beyond the statutory allocation rules, the counterfactual requires an assumption about unobserved list alliances. Unlike Leuzinger and Pukelsheim & Schuhmacher, this analysis does not carry historical or hypothetical alliances forward. List alliances are in this thesis seen as strategic and election-specific rather than fixed institutional parameters. This leads to, that election-specific, historical alliances cannot simply be assumed to persist across subsequent elections with different parties and vote distributions. In Zurich, for example, the last Kantonsrat election under Hagenbach–Bischoff took place in 2003. Assuming that the alliances formed in 2003 would still have applied in 2023 despite two decades of changes in the party system would therefore introduce a strong and unverifiable behavioural assumption. Some parties that formed list alliances in Zurich in 2003 no longer existed or contested the 2023 election. Conversely, parties such as the Green Liberal Party, which contested the 2023 election, did not yet exist in 2003. (Statistisches Amt des Kantons Zürich 2003, table 3; Statistisches Amt des Kantons Zürich 2026; Grünliberale Partei Kanton Zürich n.d.). Therefore the counterfactual preserves joint lists or list groups represented as a single unit in the observed election data. But it does not reconstruct historical alliances or assume new alliances between separately reported units. This provides a transparent counterfactual that holds the observed electoral configuration constant and estimates the mechanical allocation difference between the specified election-specific regimes.

The exclusion of unobserved list alliances is an important limitation of the counterfactual. Under Hagenbach–Bischoff, list alliances can pool votes during seat allocation and thereby improve the chances of representation for smaller lists (Pukelsheim & Schuhmacher, 2011, p. 1590). If the parties observed under Double Pukelsheim would have formed such alliances under the counterfactual HB regime, excluding them may understate smaller parties' counterfactual HB seat totals and consequently overstate their estimated gain under Double Pukelsheim. However, neither the composition of these hypothetical alliances nor the distribution of any resulting seat gains can be reliably determined. Alliance gains could accrue to larger partners or leave the aggregated representation of parties below the 10% threshold unchanged. The direction is therefore theoretically plausible but cannot be quantified uniformly across elections.

A second research-design choice concerns the electoral units and their seat magnitudes. In most cantons, the reform retained the former territorial districts or municipalities, meaning that the observed electoral units broadly correspond to those previously used under HB. The principal scope deviations are addressed in the special-cases callout below. Regarding the district magnitudes, the use of election-specific magnitudes is deliberate. District seat entitlements are generally based on population and periodically recalculated. Consequently, the magnitudes from the last pre-reform election would not necessarily have remained applicable in later elections even if Hagenbach–Bischoff had been retained. Carrying them forward would freeze an outdated distribution of seats and combine demographic changes with the effect of the allocation method. The counterfactual therefore holds each observed election's electoral units and seat magnitudes constant across the two regimes. It asks how the same

[3]These findings are based on the author's calculations of the official election data described in Appendix 1. Sources for the applicable legal provisions are provided in the preceding footnote.

votes would have been translated into seats under the former district-level rules, conditional on the contemporary distribution of seats among districts.

The remaining deviations from this general approach are summarized below.

ℹ Treatment of the Special Cases

Graubünden. The HB allocation is hypothetical and based on the former Aargau Law. Thus the observations are shown in grey and excluded from the main transition estimate.

Zug. Neuheim used Majorz before the reform. An exact Majorz counterfactual cannot be reconstructed because the results of a hypothetical second ballot are unobservable. Neuheim is therefore included using a hypothetical HB allocation.(BGE 136 I 376 E. 5.1; Kanton Zug, 2022a, 2022b, 2022c) The specific implementation approach adopted for Neuheim has no bearing on the main analysis.

Uri. Only Altdorf, Bürglen, Erstfeld, and Schattdorf used Double Pukelsheim in the elections analyzed here. They account for 37 of the Landrat's 64 seats (Standeskanzlei Uri, 2020, p. 1). Uri's vote shares, including the less-than-10% classification of smaller parties, are therefore based only on the four municipalities using Double Pukelsheim, not on the whole canton.

Valais. Unlike in the other cantons, upper apportionment is conducted separately within six electoral regions rather than canton-wide. Seats and votes are aggregated at the cantonal level only for this analysis; the actual allocation occurs independently within each region (Lüscher, 2024, p. 57).

Schaffhausen. The parliament was reduced from 80 to 60 seats when Double Pukelsheim was introduced. The counterfactual retains the 60-seat structure because comparing the two allocation methods requires the same number and distribution of seats (Regierungsrat des Kantons Schaffhausen, 2007, pp. 1-2, 6).

With the counterfactual rules and scope established, the next subsection describes the official election data and their standardization across cantons.

6.3 Data Sources and Processing

This subsection describes the official election data used in the analysis and explains how the heterogeneous cantonal source files were transformed into a harmonized dataset.

Official election results were obtained primarily from publicly available datasets published by cantonal state chancelleries and statistical offices. Where machine-readable files were not available online, the responsible cantonal authorities were contacted directly. All contacted authorities except the Canton of Schaffhausen provided data in a machine-readable format, such as CSV or XLSX; the Schaffhausen results therefore had to be extracted from official PDF documents. The precise sources and elections covered are reported in Appendix Table 1. The granularity and organization of the source data vary considerably across cantons and elections, ranging from canton-wide files to separate district- or municipality-level results. In Schwyz, for example, the 2016 results were distributed across 30 municipality-specific workbooks, whereas the 2020 and 2024 results were each provided in a single workbook. The structure of the files also differed: some, such as those for Graubünden, were prepared for

public or media use and were readily accessible, while others had originally been designed for internal administrative calculations. For most elections, information from at least two source files had to be combined.

Party and list names were standardized using readable and consistent abbreviations. The same political organization was assigned a common identifier across cantons and elections wherever possible; examples include “SVP”, “GLP”, “ALG”, and “PIRATEN”. Meaningful changes over time were nevertheless retained. In particular, “CVP” and “BDP” remain separately identified before their merger, while “DIEMITTE” is used only after the formation of the new party in 2021 (Schaub, 2022). In bilingual Valais, French- and German-language labels referring to the same party were harmonized, generally using the German abbreviation for consistency with the other cantons. However, “SP_GAUCHE_CITOYENNE” and “SP_GRUENE_UNABHAENGIGE” were retained as distinct identifiers because they represented separately reported lists competing in different linguistic regions.

The treatment of joint and affiliated lists follows the electoral units reported in the official results. Where several political organizations contested an election on a single joint list, that list was treated as one analytical unit. Examples include “SP_GRUENE” in Graubünden in 2022 and “FGA_AL_PDA” in Zurich in 2023. Such joint lists differ from list alliances under Hagenbach–Bischoff: parties in a list alliance retain separate lists but pool their votes at a particular stage of the seat-allocation process. Conversely, Schaffhausen reported several affiliated or special-interest lists associated with a broader parent party, including “SVP_KMU”,“SVP_SEN”, “JG” (Green Youth) and “JFSH” (Young Liberals) as separate electoral lists. Because these lists entered the official Double Pukelsheim allocation separately, they were also retained as separate units in the analysis. This decision faithfully reproduces the official allocation structure, although it can affect the measured number and seat share of smaller parties.

Beyond the harmonization of party and list names, the cleaning process involved restructuring vote data, identifying the relevant electoral units, and assigning their election-specific seat totals. The source files were converted into comparable tables containing party votes and the seat magnitude of each electoral district. Where parties appeared in separate columns, as in Aargau, Graubünden, and Nidwalden, these columns were reshaped into the variables “parties” and “party_votes”. Candidate-level exports required a different treatment. In three Zug and Schwyz’s 2020 election, each candidate row repeated the corresponding list’s vote and seat totals. The scripts first checked that these repeated values were consistent and then retained one observation per municipality and list, preventing the totals from being counted multiple times. Source variables such as “list_votes” and “election_mandates” were thereby converted into “party_votes” and “district_seats”. The geographic preparation depended on the structure of the official files. In Zurich, municipal party votes were summed within the electoral districts identified by the source variable “Geschaeft”. Aargau’s files already contained district totals, which were selected using the “DISTRICT” classification. Similarly, the Graubünden script extracted the existing district totals by carrying each “Wahlkreis” heading down to its corresponding “Total” row. Municipalities were retained as allocation units in Nidwalden, Schwyz, and Zug. Their names were extracted from the source tables or, for Schwyz in 2016, from the municipality-specific filenames. Uri’s script explicitly selected the workbook blocks for Altdorf, Bürglen, Erstfeld, and Schattdorf. In Valais, the 14 districts were assigned to six electoral regions in R according to the official electoral structure, preserving the

different geographic levels required for the district-level HB counterfactual and the regional DP upper apportionment. District seat magnitudes were obtained from the available official information for each election. In Aargau, district names were matched to officially sourced seat vectors or separate seat-allocation tables provided by the canton officials. Valais used explicitly recorded district magnitudes for each election year. In Zurich and Graubünden, magnitudes were reconstructed by counting elected candidates within each district. In Uri and Schwyz's 2016 election, they were calculated by summing the official DP seats within each municipality.

The shared finalization function then produced one observation per standardized party or list code and election. It created an election identifier combining the canton abbreviation and election year, such as "ZH_23", and joined the HB and DP seat totals by party. The final columns were "election_id", "parties", "vote_share", "small_party", "seats_hb", "seats_dp", "parliament_size", and "seats_allocated_dp". The last two variables distinguish the full parliament from the seats covered by the comparison: in Uri, these values are 64 and 37, respectively. District identifiers, raw party votes, and district magnitudes served as intermediate inputs rather than columns in the final combined dataset.

Vote shares were derived using the information available in each canton's official election files. In Aargau, Zurich, Zug, and Valais, party votes were divided by the corresponding district's seat magnitude to obtain voter equivalents. These quotients were rounded to the nearest integer, with half values rounded upwards, before being summed by party across districts. Each party's vote share was then calculated as its aggregated voter number divided by the total voter number of all included parties, multiplied by 100. Nidwalden instead used the voter numbers supplied in the official upper-apportionment worksheet. Schaffhausen used the published voter numbers extracted from the official result PDFs; reconstructed voter numbers were compared with these published values using election-specific tolerances to accommodate differences in reporting precision.

Other datasets supplied vote shares directly or required different denominators. In Schwyz, the 2016 shares were calculated from each party's total cantonal votes relative to the sum of votes for all retained party lists, excluding the category for ballots without a party designation. For 2020, the denominator was the total accounted votes, counted once per municipality and checked against the summed list votes. The 2024 Schwyz shares and the Graubünden shares were taken directly from the official cantonal summaries. Uri's published shares were also retained, but refer exclusively to the four municipalities included in the proportional comparison. The shared finalization function rounded all vote shares to four decimal places.

The Double Pukelsheim results were assembled separately within the canton-specific scripts from official seat tables, municipal allocations, or elected-candidate records. Valais required a separate reconstruction of the upper apportionment within its six electoral regions, followed by comparisons with published official results. Each script supplied its election-specific DP results to the shared finalization function, which joined them to the HB counterfactuals by standardized party identifier. Validation routines checked electoral-unit coverage and seat totals and, where alternative representations were available, reconciled candidate counts or aggregated lower allocations with reported list mandates or upper-apportionment totals. District-level consistency checks and final allocation totals were applied according to the available source structure. Uri's allocation checks covered the 37 proportional seats included

in the analysis, while the full parliamentary size remained recorded as 64. Unresolved HB ties were retained as alternative allocations.

All processing was implemented in the canton-specific R scripts in "data_cleaning_cantons", supported by the shared functions in "00_functions".R. Each "process_canton_*()" function combined its cleaned elections into a cantonal dataset. These nine datasets were subsequently appended row-wise using "bind_rows()" to create "all_elections", preserving one observation per standardized party or list and election. Detailed procedures, source-specific exceptions, and validation arrangements are documented in Appendix 1, with the cleaning code provided in Appendix 2.

6.4 Measures and Analytical Strategy

Here comes:

- Smaller party definition and 10% cutoff.
- Unit of analysis: election, and analyzed area
- Definition and interpretation of (_e).
- Unweighted mean used for H1.
- Seat-weighted mean as supplementary information.
- Distributional statistics reported.
- Procedure for the DP-seat-count comparison.
- Procedure for the quorum comparison.
- Procedure for examining successive elections.
- Graubünden's exclusion from the main estimate.
- why seats_allocated_dp is the denominator

For the purposes of this thesis, a smaller party is defined as an electoral list or list group with a vote share below 10% in the analyzed area. The analyzed area is equal to the whole canton except in Uri, where the vote shares cover only the four municipalities using Double Pukelsheim. Youth lists and other special-interest lists, such as women's lists or pensioners' lists, are treated as separate analytical units whenever they are reported separately in the official election results; their votes and seats are not attributed to a parent party. Accordingly, the term *party* in the analysis refers to the standardized electoral list or list group recorded in the dataset and does not necessarily correspond to a broader organizational party family. The choice of 10% as the threshold is grounded in both legal and normative considerations. The Swiss Federal Supreme Court treated 10% as an upper limit for direct quorums and a target for natural quorums when redesigning an electoral system (BGE 131 I 74 E. 5.3–5.4 ; see also Bundeskanzlei, 2013, p. 8). Using vote share rather than absolute party size is particularly appropriate because a party may be electorally small in one region while being comparatively strong in another. Low thresholds are closely linked to the principle of Erfolgswertgleichheit which is the equality in the effect of votes on seat allocation, which requires that votes should be translated into seats as equally and effectively as possible (BGE 131 I 74 E. 3.1; Bundeskanzlei, 2013, pp. 8–9). This threshold is especially relevant to the cantons analyzed here: before their reforms, each had at least one district with an electoral threshold of 10% or higher (Lüscher, 2024, p. 115).

6.5 Results

6.5.1 Main result: Smaller parties' seat-share gain

Figure 1 compares the observed Double Pukelsheim allocation with the counterfactual Hagenbach-Bischoff allocation for every cleaned election. For each election, the figure aggregates the seats and vote shares in the analyzed area of all parties below the 10% threshold defined above. The grey line connects the two allocations derived from the same election.

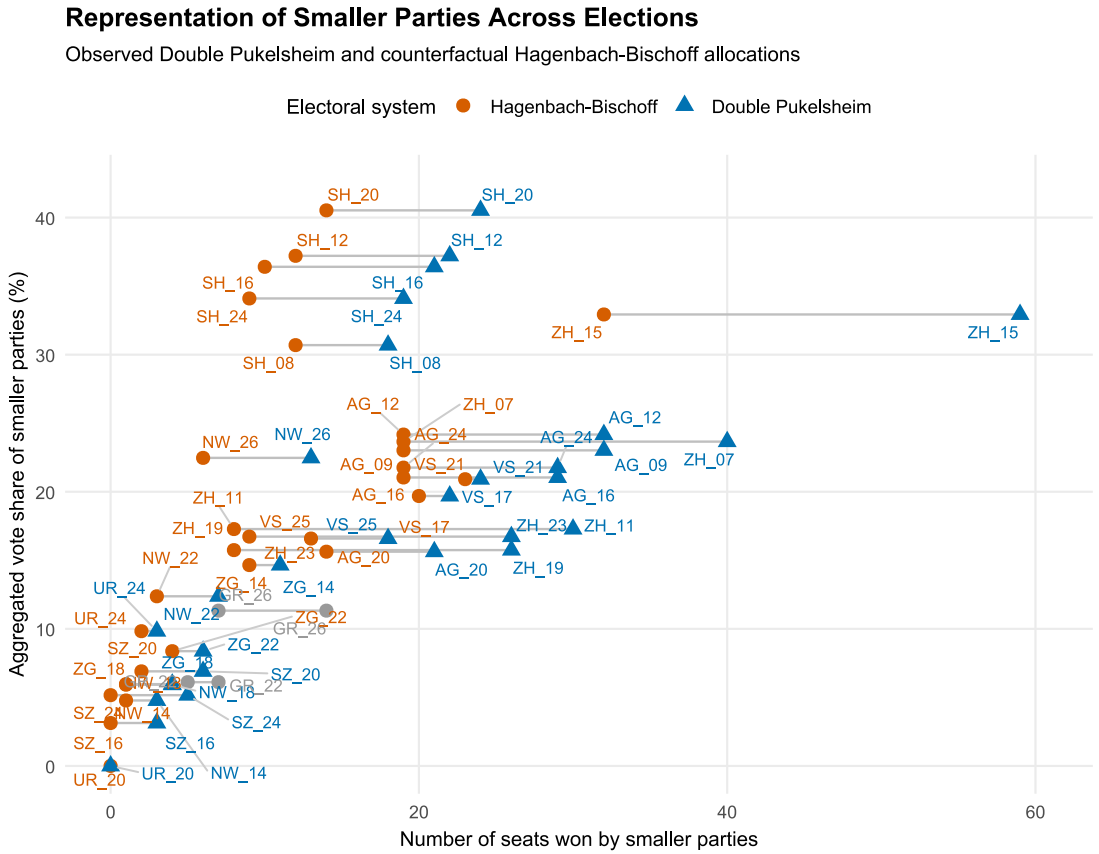


Figure 1: Seats won by smaller parties under Double Pukelsheim and the counterfactual Hagenbach-Bischoff allocation. Smaller parties have less than 10% of the vote share in the analyzed area.

6.5.2 Election-level variation

Figure 2 shows the same elections using the percentage of analyzed Double Pukelsheim seats won by smaller parties on the x-axis. Each label reports the gain under Double Pukelsheim as a change in seat share, expressed in percentage points.

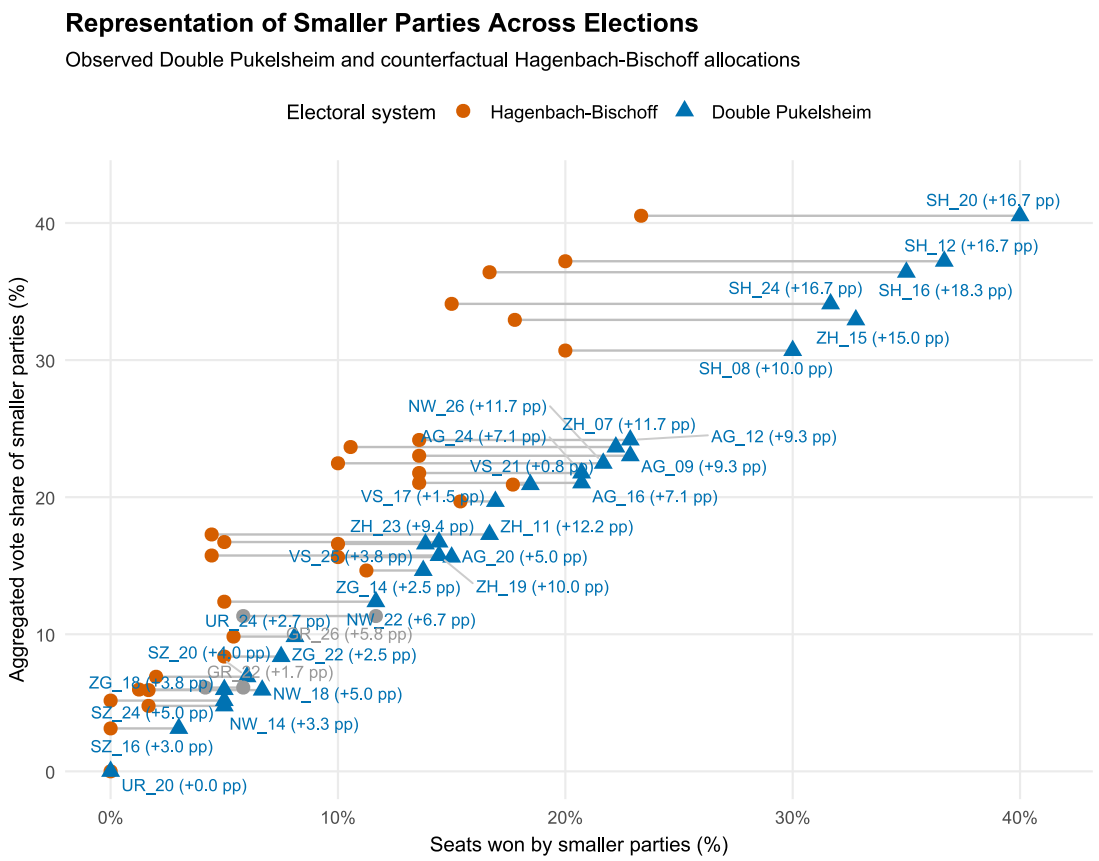


Figure 2: Percentage of analyzed seats won by smaller parties under Double Pukelsheim and counterfactual Hagenbach-Bischoff.

> **i** Hypothesis 1 summary
>
> Across all 30 elections (Graubünden excluded), the unweighted mean gain in smaller parties' share of analyzed seats was 7.7 percentage points (Q1: 3.4; median: 6.9; Q3: 11.2). The mean weighted by each election's number of analyzed seats was 7.9 percentage points. 16 individual elections exceeded five percentage points.

6.5.3 Exploratory comparison 1: Double Pukelsheim seat count

The comparison below uses the same election-level percentage-point gain as H1. It averages elections within each canton before comparing the two groups defined by the number of seats allocated through Double Pukelsheim. Graubünden is only shown for context and excluded from the group means for the same reason as in the main H1 estimate.

Table 2: Mean gain in smaller parties' share of analyzed seats by canton (percentage points). Source: author's aggregation of the official cantonal election datasets.

Canton	Group	Elections	DP seats (parliament)	Mean gain (pp)
ZH	More than 100 DP seats	5	180 (180)	11.7
AG	More than 100 DP seats	5	140 (140)	7.6
VS	More than 100 DP seats	3	130 (130)	2.1
GR	Context only (former Majorz)	2	120 (120)	3.8
SZ	At most 100 DP seats	3	100 (100)	4.0
ZG	At most 100 DP seats	3	80 (80)	2.9
NW	At most 100 DP seats	4	60 (60)	6.7
SH	At most 100 DP seats	5	60 (60)	15.7
UR	At most 100 DP seats	2	37 (64)	1.4

The mean of the canton-level gains is 7.1 percentage points in the group with more than 100 Double Pukelsheim seats and 6.1 percentage points in the group with at most 100. Their difference (more than 100 minus at most 100) is +1.0 percentage points. With only 8 cantons and distinct electoral rules, this is a descriptive comparison rather than evidence of a causal effect of the number of Double Pukelsheim seats.

7 Discussion

8 Conclusion

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10 Appendix

10.1 Appendix 1: Explanation Data Cleaning Process

The table below lists the official election-result sources used in the canton-specific cleaning scripts. It includes raw data inputs and official publications used to reconstruct or validate reported results; legal and methodological sources used only to specify the counterfactual allocation rules are excluded.

Canton	Data source
Aargau	Statistisches Amt Kanton Aargau, **Grossratswahlen: Detailresultate, ab 1985** (elections 2009, 2012, 2016, 2020, and 2024); Department für Finanzen und Ressourcen des Kantons Aargau (2009) and Regierungsrat des Kantons Aargau (2013) for validation of the 2009 and 2012 cantonal seat totals; Staatskanzlei des Kantons Aargau (2012, 2016) for the district seat magnitudes.
Graubünden	Kanton Graubünden, **Pressedaten Proporzwahl kantonal: Grossratswahlen** (official Excel workbooks for 2022 and 2026).
Nidwalden	Staatskanzlei Nidwalden, **Wahlergebnis der Landratswahlen** (unpublished Excel workbooks for 2014, 2018, 2022, and 2026, supplied by the Kanzleisekretariat via email).
Schaffhausen	Kanton Schaffhausen, **Frühere Urnengänge (seit 2003)** (official result PDFs for 2008, 2012, and 2016); Staatskanzlei Kanton Schaffhausen, **Resultate der Kantonsratswahl** (official result PDFs for 2020 and 2024).
Schwyz	Staatskanzlei Kanton Schwyz, unpublished raw data supplied via email: municipality-level CSV exports (30 files for 2016), a candidate-result CSV export (2020), and an Excel workbook (2024).
Uri	Standeskanzlei Uri, **AuswertungKapitel2-2024** (unpublished Excel workbook containing the 2020 and 2024 results, supplied via email); **Landratswahlen 2024: Statistische Auswertung** for party-label validation.
Valais	Staat Wallis, official district-level Grand Council result workbooks (2017, 2021, and 2025); official final results for 2017; Grand Council validation-commission reports for the constitutive sessions of April 2021 and April 2025; Kanton Wallis publications on district seat magnitudes for 2017, 2021, and 2025.
Zug	Kanton Zug, official open-data candidate-result CSV exports for the 2014, 2018, and 2022 cantonal elections.
Zürich	Amt für Statistik und Daten Kanton Zürich, **Archiv Wahlresultate Kantonsratswahlen** (official party-result and candidate-result Excel workbooks for 2007, 2011, 2015, 2019, and 2023).

Table 1: Official election data sources used in the canton-specific data-cleaning scripts.

10.2 Appendix 2: Code for Data Cleaning

10.3 Declaration of AI-Assistance